

CMPDI DETAILED PROJECT REPORT (DPR) - Rajmahal Opencast ECL

Subsidiary: ECL | Capacity: 17.0 MT | Geological Basin: Rajmahal Basin, Jharkhand

Official Publication | Published by Ministry of Coal & CMPDI Ranchi | Date: 11 September 2026

This Detailed Project Report (DPR) formulates the mine planning, equipment sizing, and geological stripping layout for Rajmahal Opencast ECL under ECL.

Geological Reserve & Setting: Situated in the Rajmahal Basin, Jharkhand, the mine exploits thick coal seams belonging to the Barakar Formation. Total geological reserves are estimated at 480 Million Tonnes.

Rated Production Capacity: 17.0 MT under mechanized Opencast operations. Stripping ratio is projected at 2.80 m3/t, utilizing 42 m3 electric rope shovels, 240T rear dump trucks, and high-capacity bucket wheel systems.

Environmental Mitigation & FMC: Coal evacuation is planned via a First Mile Connectivity (FMC) rapid loading silo with in-pit crushing and a 4.2 km overland conveyor to eliminate surface haul truck emissions.

Technical Parameter	Project Specification	DGMS / Ministry Compliance
Subsidiary / Mine Name	ECL - Rajmahal_Opencast_EC	Approved by CIL Board
Rated Annual Capacity	17.0 MT	Environmental Clearance Granted
Mining Method	Opencast	High-wall Stability Factor > 1.3
Stripping Ratio	2.80 m3/t	Continuous Reclamation Mandated
Evacuation Mode	Rapid Loading Silo / FMC	Zero Surface Dust Spillage